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Artificial neural networks applied to cancer detection in a breast screening programme

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ABSTRACT

Breast screening is a method of detecting breast cancer at a very early stage. The first step involves taking an X-ray, called a mammogram, of each breast. The mammogram can detect small changes in breast tissue which may indicate cancers which are too small to be felt either by the woman herself or by a doctor.

The World Health Organisation's International Agency for Research on Cancer (IARC) concluded that mammography screening for breast cancer reduces mortality. This means that out of every 500 women screened, one life will be saved.

The present research uses the information obtained from the breast screening programme carried out in the public health area of Aviles (Principality of Asturias, Spain) from 1999 to 2007. The public health area of Aviles is formed by nine municipalities with a total of 160,000 inhabitants. The selection of the public health area was based on the following criteria:

- This is the first screening programme performed in the area.
- Almost 100% of the population in the area benefit from the public health system.
- The Aviles public health area is a well-defined area of the region that does not send patients to other public health areas, which makes the study easier and more accurate.

This paper describes a neural network based approach to breast cancer diagnosis; the model developed is able to determine which women are more likely to suffer from a particular kind of tumour before they undergo a mammography.

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1. Introduction

Breast cancer is currently one of the most common diseases among women in the developed world. In the Principality of Asturias (Spain), one in every 500 women will develop breast cancer at some time in her life [1]. In terms of mortality, it is the leading cause of death among women aged between 35 and 64 years, and is the leading cancer-related cause of death in the female population as a whole [2,3].

The efficacy of breast cancer screening using mammography has been studied extensively since the mid-1960s. The meta-analyses and reviews of the controlled trials conducted in the United States [4,5], in Sweden [6,7], Canada [8], and the United Kingdom [9] have shown that mammography is clearly effective in women over the age of 50, and of limited efficacy

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in those aged under 50. In the majority of these studies, the various findings indicate that screening women from age 50 onwards may lead to a reduction in mortality of 20%–35% after 10 years. As a consequence of these findings, the majority of scientific and professional organizations recommend mammographic screening, and since the late 1980s a large number of countries have implemented population-screening programmes. However, it was not clear if this effect would persist outside the controlled trial environment or be enhanced [10].

In developed countries the number of women reaching the age for screening programmes is increasing, which implies an increase in medical costs.

In spite of the fact that the early detection of women with breast cancer is vital, the minimization of diagnostic expense is essential and therefore screening programmes should be optimized in order to improve their efficiency [10].

In any case, these screening programmes have been providing the medical community with increasingly reliable and accessible amounts of data, which suggests the need for computational tools to interpret such data and to organize it efficiently.

Clustering techniques are largely recognized as being useful exploratory tools for breast cancer data analysis [11–14]. Women showing similar expression patterns over a wide range of experimental conditions can be clustered together. This relies on the hypothesis that women that belong to the same cluster are co-regulated and involved in related functions [15]. Nevertheless, clustering algorithms still have limits, particularly for the estimation of the number of clusters and the interpretation of hierarchical dendrograms, which may significantly influence the output of the analysis process.

The present article introduces a SOM based clustering algorithm for the pre-processing of a wide sample of information from a breast cancer screening programme. Once clustered, the above-mentioned information will be used as input for a pattern classification technique such as support vector machines (SVM) in order to implement a model that allows for a reduction in the number of subjects that have to undergo the screening programme.

2. Materials and methods

2.1. Self-Organizing Maps (SOM)

The self-organizing map (or Kohonen map) as proposed in [16] and discussed in [17] provides an easily-implemented algorithm for dimension reduction and visualization in the field of unsupervised learning. The data vectors are projected onto positions on a two-dimensional grid. The grid consists of a set of $k \times k$ regular ordered nodes $n^{(l)}$, $l = 1, \ldots, L = k^2$, each associated with a prototype vector $u^{(l)}$ that is of the same dimension as the data points $\{x\}_\lambda$. The training scheme of the SOM is similar to the k-means clustering with two important complementary rules. Firstly, the two-dimensional grid positions $n^{(l)}$ source remains unchanged. Secondly, in each learning step, not only the winner node's $u^{(k)}$ source is updated, but also its neighbors' prototype vectors (neighboring according to the two-dimensional position in the grid). The learning step width e for updating neighbors decreases over grid distance, expressed by a Gaussian weight function with width σ. During training, the parameters e and σ are decreased. The critical parameters of the SOM are the number of nodes L, the initial and final neighborhood size (σ_i, σ_f) and the number of iterations ne. In previous researches SOM have been applied successfully to the classification of XRM (X-ray mammography) based and biopsy data [18,19].

2.2. Support vector machine

In recent years, kernel based methods have been the subject of much interest and research in the machine learning community. The success of many kernel methods is based on the concept of combining well-known linear algorithms such as Principal Component Analysis, with non-linear kernel functions. While the application of these functions allows for more powerful non-linear solutions, the kernelized algorithms [20,21] retain most of the properties of their linear versions. The most prominent algorithm among these is the support vector machine (SVM) proposed by Vapnik [22] for binary classification.

The SVM is gaining popularity due to many attractive features and its great classification performance, demonstrated in numerous applications. It realizes pattern recognition between two classes by finding a decision function (hyperplane) determined by selected points from the training data, termed support vectors (SVs). In general, this hyperplane corresponds to a non-linear decision boundary in the input space. While traditional techniques for pattern recognition are based on minimizing the empirical risk, i.e. on the attempt to optimize the performance on the training set, SVMs minimize the structural risk, i.e. the probability of yet-to-be-seen patterns being classified correctly for a fixed but unknown probability distribution of the data.

The SVM with a Gaussian kernel is trained and its parameters σ (width of Gaussian) and C (regularization) are optimized using the LIBSVM3 package, which combines the SMO [23,24] and the SVMLight4 algorithm.

2.3. Multivariate adaptive regression splines

MARS was first proposed by Friedman [25] as a flexible procedure which models relationships that are nearly additive or involve interactions with few variables. The modelling procedure is inspired by the recursive partitioning technique

Table 1
Confusion matrix for binary classification.

Class/recognized	As positive	As negative
Positive	*tp*	*fn*
Negative	*fp*	*tn*

governing classification and regression trees [26] and generalized additive modelling [27], resulting in a model that is continuous, with continuous derivatives. It excels at finding optimal variable transformations and interactions, the complex data structure that often hides in high-dimensional data.

MARS essentially builds flexible models by fitting piecewise linear regressions; that is, the nonlinearity of a model is approximated through the use of separate regression slopes in distinct intervals of the predictor variable space. Therefore the slope of the regression line is allowed to change from one interval to the other as the two 'knot' points are crossed. The variables to use and the end points of the intervals for each variable are found via a fast but intensive search procedure. In addition to searching variables one by one, MARS also searches for interactions between variables, allowing any degree of interaction to be considered.

The general MARS function can be represented using the following equation [25]:

$$\hat{f}(x) = a_0 + \sum_{m=1}^{M} a_m \prod_{k=1}^{K_m} \left[s_{km}(x_{v(k,m)} - t_{km}) \right]_+$$

where a_0 and a_m are parameters, M is the number of basis functions, K_m is the number of knots, σ takes on values of either 1 or -1 and indicates the right/left sense of the associated step function, $v(k, m)$ is the label of the independent variable, and t_{km} indicates the knot location.

The optimal MARS model is selected in a two-stage process. Firstly, MARS constructs a very large number of basis functions which are selected to overfit the data initially, where variables are allowed to enter as continuous, categorical, or ordinal – the formal mechanism by which variable intervals are defined – and they can interact with each other or be restricted to only enter as additive components. In the second stage, basis functions are deleted in order of least contribution using the generalized cross-validation (GCV) criterion [28]. A measure of variable importance can be assessed by observing the decrease in the calculated GCV values when a variable is removed from the model. The GCV can be expressed as follows:

$$LOF(\hat{f}_M) = GCV(M) = \frac{1}{N} \sum_{i=1}^{N} [y_i - \hat{f}_M(x_i)]^2 \frac{1}{\left[1 - \frac{C(M)}{N}\right]^2}$$

where there are N observations, and $C(M)$ is the cost-penalty measures of a model containing M basis function (therefore the numerator measures the lack of fit on the M basis function model $f_M(x_i)$ and the denominator denotes the penalty for model complexity $C(M)$. MARS has been widely used in handling problems in the areas of forecasting and classifications [29–34].

3. Performance measures

In a scenario where people are tested for a disease, the outcome test can be positive (sick) or negative (healthy), while the actual health status of the person may be different. Sensitivity and specificity are statistical measures of the performance of a binary classification test. Sensitivity measures the proportion of actual positives, which are correctly identified, and specificity measures the proportion of negatives, which are correctly identified.

$$\text{Sensitivity} = \frac{tp}{tp + fn}$$

$$\text{Specificity} = \frac{tn}{fp + tn}.$$

Supervised Machine Learning (ML) has several ways of evaluating the performance of learning algorithms and the classifiers they produce. Measures of the quality of classification are built from a confusion matrix which records correctly and incorrectly recognized examples for each class. Table 1 presents a confusion matrix for binary classification, where *tp* are true positive, *fp* false positive, *fn* false negative, and *tn* true negative counts.

Accuracy assesses the overall effectiveness of the algorithm. It is given by

$$\text{Accuracy} = \frac{tp + tn}{tp + fp + fn + tn}.$$

The avoidance of failure complements accuracy, or the ability to correctly label examples. Youden's index (γ) [35] evaluates the algorithm's ability to avoid failure.

$$\gamma = \frac{\text{Sensitivity}}{(1 - \text{Specificity})}.$$

Table 2
Variables analyzed by the surgeons during the screening programme.

Common variables		Postmenopause women	Women with pregnancies
Age	Horm antic.	Age menarche	Age first pregnancy
Family antec.	Years horm antic.	Age menopause	Number of pregnancies
Prev. mammograph R	Prot. L	Estrogenic batch	Number of lactations
Prev. mammograph L	Prot. R	Kind of menopause	Months of lactation
THS	Surgery L		
Years THS	Surgery R		

4. Experimental results

The present research uses the information obtained from the breast cancer screening programme carried out in the public health area of Aviles (Principality of Asturias, Spain) from 1999 to 2007. The public health area of Aviles is formed by nine municipalities with a total of 160,000 inhabitants.

The number of women sampled during the Aviles screening programme was 49,501. A breast tumour was identified in 100 of the women analyzed due to the screening programme, which represents 0.2% of the total population. 20 variables were analyzed and these are listed in Table 2.

4.1. Data pre-processing

Given the particular characteristics of the database used in the modelling process, where only 0.2% of the sample suffer the disease (100 individuals with cancer versus 49,401 healthy individuals), pre-processing of the information is necessary before a data based modelling can be undertaken, as is the case of the neural network model.

In spite of the number of variables taken into account in the screening programme (20), there are unsampled factors that influence the appearance of the disease. For every sick patient it is possible to find a high number of healthy patients with very similar values in the variables. In order to solve this problem, the negative influence that the large amount of healthy patients have on the small sample of sick ones must be avoided, so it is necessary to perform a dimensional reduction of the sample for the particular case of healthy patients.

What is sought by the above-mentioned dimensional reduction is a selective group of healthy patients of similar characteristics inside a set identified by a single individual who represents all the others. After the dimensional reduction of the sample we will have a new set of healthy women smaller than the original but representative of the population.

Many technologies allow for sample clustering [36,37], of which the Self-Organizing Map (SOM) is the most outstanding [38]. A self-organizing map is a type of artificial neural network that is trained using unsupervised learning to produce a low-dimensional (typically two-dimensional), discrete representation of the input space of the training samples called a map, the purpose of which is to discover the underlying structure of the information in it.

A self-organizing map consists of components called nodes or neurons. Associated with each node is a weight vector of the same dimension as the input data vectors and a position in the map space. The usual arrangement of nodes is a regular spacing in a hexagonal or rectangular grid. The self-organizing map describes a mapping from a higher dimensional input space to a lower dimensional map space.

In order to locate an individual in a determined subset, the self-organized maps utilize competitive learning. When a training example is fed to the network, its Euclidean distance to all weight vectors is computed. The neuron with weight vector most similar to the input is called the best matching unit (BMU). The weights of the BMU and neurons close to it in the SOM lattice are adjusted towards the input vector. The magnitude of the change decreases with time, and with distance from the BMU. This process is repeated for each input vector for a large number of cycles (1000 cycles in this case). The network winds up associating output nodes with groups or patterns in the input data set.

After the network training, the input information will be grouped in a series of sets equal to the dimension of the network (or the number of neurons used in the training, which is the same thing) and every group will come unequivocally identified by the weight vector of the neuron, which is no more than a hypothetical patient with typical values in the 20 variables that define it, the values of these variables being those that best define the behaviour of the set that it represents.

The problem at this moment is to identify the most suitable number of subsets in which the healthy population can be classified. As was explained before, the 49,401 healthy individuals forming the population will be substituted by the director vectors of each of the subsets, creating a new reduced population of healthy women. This reduced subset is representative of the original population.

Therefore, the subset must be large enough in order to represent the variability of the healthy population but should be reduced enough in order to obtain a balanced sample of healthy individuals represented by their director vectors versus sick women represented by each individual of the original sample.

As the ideal number of subsets is unknown, an iterative procedure was developed. This procedure is as follows: several SOM networks of increasing dimensions were trained with the original set of healthy women (49,401 patients). Table 3 shows the dimensions of the used networks, as well as the number of neurons of every dimension and therefore the number of finally classified groups.

Table 3
Dimension of the SOM networks trained.

SOM dimension	Number of neurons	Weight vectors
6 × 6	36	36
8 × 8	64	64
10 × 10	100	100
11 × 11	121	121
12 × 12	144	144
13 × 13	169	169
14 × 14	196	196
15 × 15	225	225
22 × 22	484	484

Table 4
Specificity and sensibility of the MARS models.

Vec Som	Cancer patients	MARS models	True healthy	False sick	True sick	False healthy	Specifi. (%)	Sensibi. (%)
36	100	Model 1	931	48,470	100	0	01.9	100.0
64	100	Model 2	16,836	32,665	91	9	34.0	91.0
100	100	Model 3	27,163	22,338	74	26	54.9	74.0
121	100	model 4	27,397	22,104	73	27	55.3	73.0
144	100	Model 5	34,176	15,325	63	37	69.0	63.0
169	100	Model 6	35,140	14,361	63	37	71.0	63.0
196	100	Model 7	35,772	13,729	61	39	72.3	61.0
225	100	Model 8	42,936	6,565	43	57	86.7	43.0
484	100	Model 9	46,780	2,721	28	72	94.5	28.0

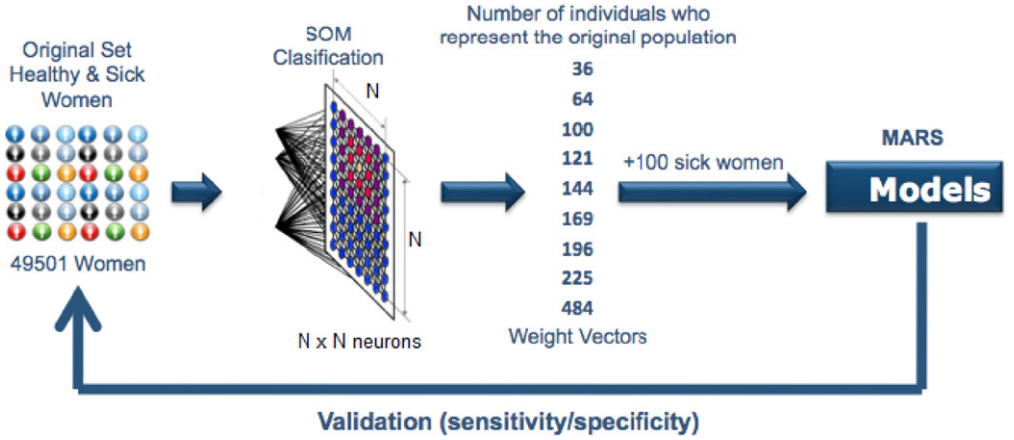


Fig. 1. The classification process and set validation of the original population.

Afterwards, an algorithm of unsupervised learning and low computation cost (MARS) was used in order to train a different model for every SOM network trained, or what is the same, for every set of weight vectors. Therefore, the final input used to train every MARS model will be the result of adding 100 patients with the pathology to the corresponding number of healthy patients from the classification SOM. The fact of not directly using a neuronal network to train every model and using MARS algorithm instead is determined by the minor resources consumption and the minor computation time of using the same ones [39].

Finally, every MARS model trained will be validated with the initial set of information (49,501 patients) evaluating the sensitivity and specificity shown. The whole process chosen to identify the number of weight vectors has ensured a better fitting to subsequently train a neuronal network with the above-mentioned sets of information of small dimension. The results obtained can be observed in the Table 4, and a schematic layout of the classification process can be observed in Fig. 1.

It may be observed in Table 4 that as the specificity of the model increases its sensitivity diminishes, which is to say that as the set of healthy patients increases, so the proportion of sick patients diminishes relatively; the models tend to identify more sick patients as healthy. The reverse is also true.

In order to determine the adequate size of the reduced sample of healthy patients, the selection criterion has been to use those groups which have obtained a specificity and sensitivity over 60% in the validation process, which reduces the final set to three groups of healthy patients in 144, 169 and 196 individuals.

Table 5
Arrangement for importance of the MARS algorithm.

Variable	*n* subsets	Gcv	Rss
Age	39	−74.04	100.00
Age first pregnancy	38	−84.10	85.30
Months of lactation	38	−84.10	85.30
Number of lactations	37	−91.49	73.74
Age menopause	37	−91.49	73.74
Number of pregnancies	36	−100.00	61.35
Family antec.	35	−98.28	59.48
Age menarche	24	−82.33	34.79

Gcv: Generalized Cross Validation (GCV) of the model (summed over all responses). The GCV is calculated using the penalty argument. Rss: Residual sum-of-squares (RSS) of the model (summed over all responses).

Table 6
Confusion matrix for the three different SVM models (first validation).

SVM model	Training set	*tp*	*tn*	*fp*	*fn*
Model 5	144 + 100	51	39,688	9,713	49
Model 6	169 + 100	99	30,129	19,272	1
Model 7	196 + 100	100	34,109	15,328	0

Table 7
Performance measures (first validation).

SVM model	Training set	Specificity	Sensitivity	Accuracy	Youden's index
Model 5	144 + 100	0.8034	0.51	0.8028	0.3134
Model 6	169 + 100	0.6099	0.99	0.6107	0.5999
Model 7	196 + 100	0.6886	1	0.6893	0.6886

4.2. Training and testing the SVM model

It turns out that it is necessary to define the set of input variables with which the SVM will be trained. This is due to the fact that the larger the number of variables, the more data is produced and the more computing resources are needed. It must be remarked that not all the variables contribute the same amount of information to the final mode; in many cases the contribution is nil and therefore the variables can be withdrawn from the model.

An additional advantage of the use of the MARS algorithm for the validation of the dimensional reduction with the SOM networks is that the MARS training algorithm allows for the identification and arrangement for the importance of the relevant variables that have contributed information for the construction of the same model. The arrangement by importance that the MARS algorithm realizes for the three models considered suitable (144 + 100, 169 + 100 and 196 + 100 individuals) can be seen in Table 5.

In order to evaluate the result, three-fold cross validation was used. A program was written in C to divide the data randomly into three different sets for training and testing the classifier. The training data included the eight feature attributes (input variables) previously selected (Table 5) and a class attribute (breast cancer detected) while the test data had only the eight features attribute and excluded the class attribute.

SVM Torch, a freely available C++ based object-oriented machine learning library, was used for training and testing the model [40]. The polynomial kernel of the SVM was trained using all the instances (244, 269 and 296 respectively) and tested with the original sample (49,501 instances) for evaluating the classifier's effectiveness in the first instance. A final validation was performed with data from the current year (3852 new women).

4.3. Performance evaluation

The classifier output for the first test data (the original set of 49,501 women) was compared with the original class attribute for identifying true positives, true negatives, false positives and false negative values. Table 6 gives these values in the form of a confusion matrix.

This confusion matrix was used to arrive at the various metrics discussed in Section 3. The performance measures were calculated individually for the different models, and their value is given in Table 7.

Accuracy approximates how effective the algorithm is by showing the probability of the true value of the class label. Table 7 and Fig. 2 shows that the accuracy of model 5 is better (80.28%) than the other two models (61.07% and 68.93% respectively).

Sensitivity and specificity approximate the probability of the positive and negative label being true (assesses the effectiveness of the algorithm on a single class). Referring to Table 7 and Fig. 2 it can be observed that the specificity of

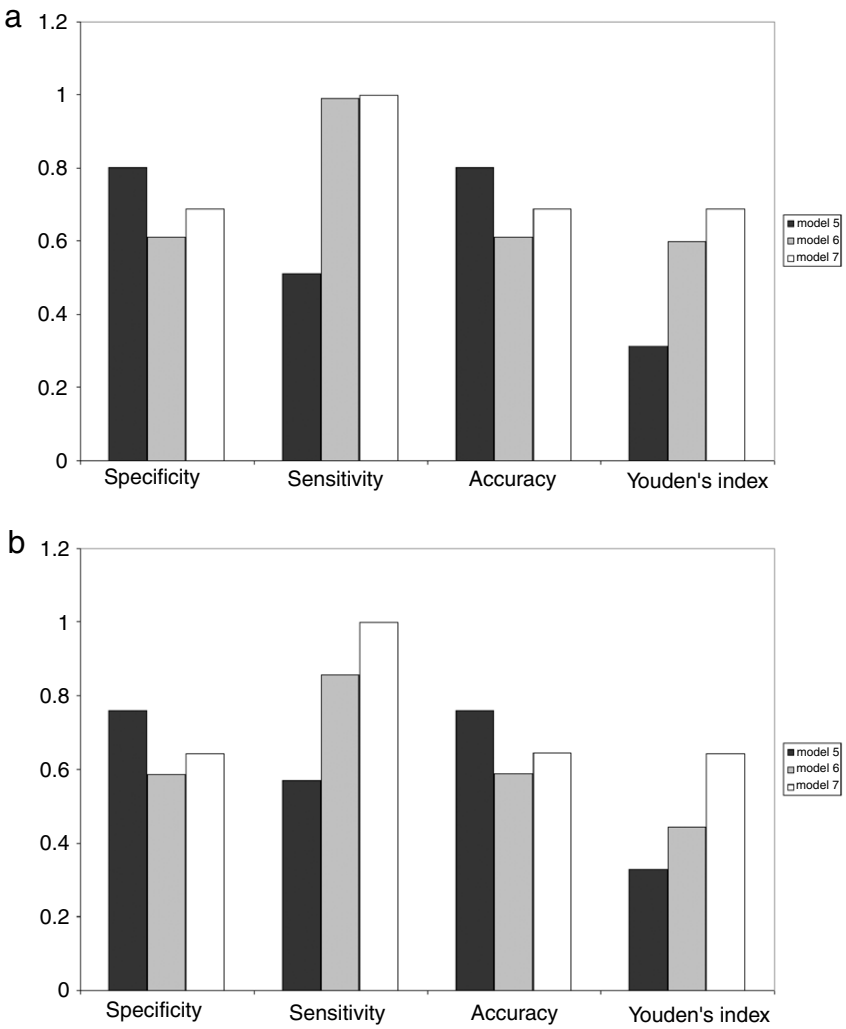


Fig. 2. Performance of SVM models for measures (Accuracy, Sensitivity, Specificity, Youden's index). (a) First validation. (b) Second validation.

the model 5 is better than the other two models. However, the sensitivity of the model 7 is higher with 100% indicating that it succeeded in identifying all the true positives correctly. The sensitivity of the model 6 is 99%.

Youden's index has been traditionally used to compare diagnostic (medical) abilities of two tests. A higher value of Youden Index (γ) means a better ability to avoid failure. Referring to Table 7 and Fig. 2 it is seen that model 7 gave a value of 0.6886, while on the other hand model 5 gave a poor value of 0.3134. This implies that model 7 is better at arriving at a correct diagnosis (correct classification) compared to that of models 5 and 6.

In the second validation the classifier output for the current year data (3852 new women) was compared with the original class attribute for identifying true positives, true negatives, false positives and false negative values. Table 8 gives these values in the form of a confusion matrix.

The performance measures were again calculated individually for the different models and their value is given in Table 9.

Table 7 and Fig. 2 shows that the best values of sensitivity by Youden's index were obtained for model 7 and the best accuracy is again obtained for the model 5.

5. Conclusions

In this paper, we have examined the usage of the polynomial kernel of Support Vector Machines (SVM) in actual clinical diagnosis of breast cancer. The performance of these models was investigated for classifying breast cancer. Known sets of cytologically proven tumour data obtained from Aviles Hospital were used to train the model to categorize cancer patients according to their diagnosis.

In addition to the usual performance measures, other metrics used in medical diagnosis, such as Youden's index, were also evaluated.

Table 8
Confusion matrix for the three different SVM models (second validation).

SVM model	Training set	*tp*	*tn*	*fp*	*fn*
Model 5	144 + 100	4	2922	923	3
Model 6	169 + 100	6	2268	1577	1
Model 7	196 + 100	7	2480	1365	0

Table 9
Performance measures (second validation).

SVM model	Specificity	Sensitivity	Accuracy	Youden's index
Model 5	0.7599	0.5714	0.7596	0.3313
Model 6	0.5898	0.8571	0.5903	0.4469
Model 7	0.6449	1	0.6456	0.6449

After the second validation, performed with 3852 new cases used neither for the training of the model nor for the first validation, it was determined that the most suitable model was model 7, in which 100% of the patients with a high probability of suffering the disease will be identified correctly (see Table 8). In addition over the 64% of the patients with low possibilities of suffering from the disease will also be identified correctly.

The main conclusion in view of these results is that the application of the screening programme to only the 36% of the population will lead to the identification of the same number of cases if the population has been previously filtered with the classification model. Therefore this work indicates that SVM can be effectively used for breast cancer diagnosis to help oncologists in order to decide which women should participate in a screening programme and which should not.

Even though some progress has been achieved, there are still many challenges remaining and directions for further research, such as developing better classification algorithms and integration of classifiers to reduce false positives.

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